

Response: Dabrowski Hilbert–Smith preprint

Sherlock reverse-verification notes

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Independent read of Integral tail signatures and the Hilbert–Smith conjecture

Upstream: github.com/adbrw/HS_proof SHA: abb0c44...

PDF SHA-256: e305fd47... Repo: github.com/ajjcoppola/hilberts-5th-oai

Original questions

1. Overview: how faithful p -adic actions are excluded near the identity homeomorphism.
2. Geometric argument, or algebraic invariants in every Euclidean neighborhood?
3. Public / internet responses to this AI-generated proof?
4. Categorize the proof; what does it say about the structure of $\mathbb{R}^n$?
5. Hold/break under Menger-type sponges as $\mathbb{R}^n$ replacements (where the theorem is false)?
6. Is the contradiction that $\mathbb{R}^n$ cannot factor through an orbit space with “wonky” local sphere maps?

TL;DR

Dabrowski’s preprint claims a proof of the Hilbert–Smith conjecture by ruling out effective continuous $\mathbb{Z}_p$ -actions on connected finite-dimensional manifolds (boundary allowed). The argument is not Yang’s orbit-dimension obstruction and is not Pardon’s three-manifold mapping-class argument. It is an integrality-versus- p -divisibility contradiction for an *integer signature tail* in ultimate quadratic L_4 : a small open tail of $\mathbb{Z}_p$ Haar-averages a degree-one pinch in a Euclidean chart; finite quotients and a character detector produce a duality-compatible homotopy idempotent on $S^n \times \mathbb{CP}^2$; iterated Karoubi localization boundaries recover one copy of $\mathbb{CP}^2$, giving a class whose ordinary signatures are eventually 1. Independently, on a fixed compact free C_p -space, permutation-tensoring is locally p copies of the identity and is nilpotent after Mayer–Vietoris, so every L_4 class has eventually p -divisible signatures. Hence $1 \in p\mathbb{Z}$. This is the same obstruction family as the OpenAI (2026-09-23) Witt-sheaf argument, with different packaging. A targeted audit found no outright failure in the load-bearing PDF lemmas; several steps remain weak (negative K / tail extraction, nilpotence of ν , the graph corner, $\mathbb{CP}^2$ evaluation, and an independently printed Lean axiom list). On Menger-type sponges, where faithful $\mathbb{Z}_p$ actions exist, the writeup correctly refuses to run because chart and combinatorial dual-cell hypotheses fail. The claimed Lean 4 certificate was not rerun in this session.

Proof overview (Q1, Q2, Q4, Q6)

Proof overview — Dabrowski Hilbert–Smith preprint (2026-10-08)

Answers to reverse-verification questions Q1, Q2, Q4, Q6, applied to A. Dabrowski’s *Integral tail signatures and the Hilbert–Smith conjecture*. Source: vendored PDF in `paper/HSproof.pdf`.

Statement

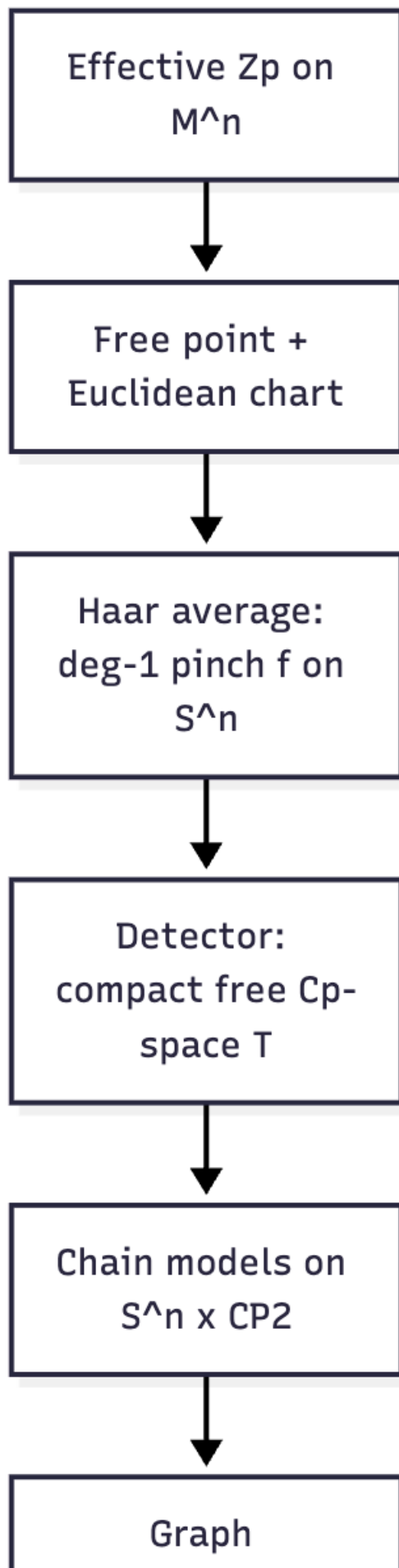
Theorem 1.1 (p-adic exclusion). For every prime p , a continuous action of $\mathbb{Z}_p$ on a connected finite-dimensional manifold has nontrivial kernel.

Corollary 1.2 (Hilbert–Smith). A second-countable locally compact Hausdorff group acting continuously and effectively on such a manifold is a Lie group with its given topology.

Manifolds are Hausdorff and second countable, and **may have boundary**. All actions are jointly continuous. Corollary 1.2 is the standard reduction: locally compact structure + Newman ([Pa19]) + Theorem 1.1 + Gleason–Yamabe (no small subgroups $\Rightarrow$ Lie). The Lean `Challenge.lean` states the **boundaryless** form; the paper recovers the boundary case by restricting to the interior (homeomorphisms preserve the interior by local homology).

Mechanism in one paragraph

Assume an effective $\mathbb{Z}_p$ -action on a connected n -manifold, $n \geq 1$. A free point exists (Lemma 8.1). Center a Euclidean chart; Haar-average coordinates of a small open tail $H = p^j \mathbb{Z}_p$ to get an invariant degree-one pinch $f : S^n \rightarrow S^n$. A character detector embeds a compact free C_p -space T (odd sphere) from the free covering Z^+/H' . On the product Poincaré manifold $W = S^n \times \mathbb{CP}^2$, shrinking triangulations supply controlled chain models; finite quotients $G_i = H/K_i$ act up to homotopy. Compression to an exterior band, then a graph idempotent E , splits in Karoubi completion to a Poincaré corner $d \in L_{n+4}(B_{G,Z}(T \times S^n))$. Forgetting equivariance recovers the scalar germ of f . Iterated Karoubi localization boundaries Δ_T cut the sphere down to a point and retain one copy of $\mathbb{CP}^2$, giving $\alpha \in L_4(A_G(T))$ with **signature tail constantly** 1 (Proposition 13.1). Independently, on any compact free C_p -space T , tensoring with the positive permutation form $\tau_i = \mathbb{Q}[G_i/P_i]$ is locally p copies of the identity; controlled Mayer–Vietoris makes $\nu = p \cdot \text{id} - \tau \otimes (-)$ **nilpotent**. Equivariant signature characters plus cyclotomic arithmetic force every class in $L_4(A_G(T))$ to have **eventually p -divisible ordinary signatures** (Theorem 4.3). Hence $1 \in p\mathbb{Z}$.



Q1 — Why no p-adic homeomorphisms near id?

Same geometric reason as OpenAI, packaged differently.

1. **Small open tails.** Effectivity plus Roberts's theorem on pointwise-finite families produce a free point (Lemma 8.1). Joint continuity then supplies $H = p^j \mathbb{Z}_p$ with arbitrarily small Euclidean displacements on a chart ball. Finer kernels $K_i = p^{\ell_i} \mathbb{Z}_p$, $\ell_i \rightarrow \infty$, give finite cyclic groups $G_i = H/K_i$ of unbounded p -power order.
2. **Averaging near id.** The Haar coordinate average $F_0(x) = \int_H hx d\mu(h)$ is H -invariant and C^0 -close to the identity on a chart, so the compactified pinch f has degree one. Finite groups have no smaller open subgroup to average over while keeping degree one.
3. **Arithmetic uses the sequence, not a single cover.** Divisibility is a statement about the **tail** of $L_4(A_G(T))$: signatures of the coordinate forms (V_i, b_i) are eventually divisible by p . Realization produces the constant tail 1. That needs infinitely many finer G_i , which only $\mathbb{Z}_p$ supplies.

So the disproof is: a faithful action near the identity would manufacture a Poincaré class over a fixed free C_p -space whose ordinary signatures stay 1, while permutation-tensor nilpotence forces those signatures into $p\mathbb{Z}$.

Q2 — Geometric or algebraic?

Mostly algebraic L-theory, with a thin Euclidean shell.

Layer	Nature	Content
Chart / free point / Haar pinch	Geometric	Local Euclidean coordinates, Newman in the corollary, orientation of $S^n \times \mathbb{CP}^2$
Detector / sheets / triangulations	Topological / PL	Equivariant map $Z^+ \rightarrow T = S^{2m-1}$; mesh $\eta_i \rightarrow 0$ chain approximations
Duality / corners / localization	Algebraic	Ultimate lower quadratic L -theory $L_k = \pi_k L^{(-\infty)}$; Karoubi filtrations [CP95]; Ranicki Poincaré pairs [Ran80I, Ran80II]; Balmer–Schlichting splitting [BS01]
Contradiction	Arithmetic	Integer signature tail: $1 \in p\mathbb{Z}$

The obstruction is **not** Yang/Bredon–Raymond–Williams cohomological-dimension raising. Orbit-space dimension never appears. The invariant that must hold in every Euclidean neighborhood is an **integer signature** realized by $\mathbb{CP}^2$ after forgetting a controlled equivariant corner.

Q4 — What the proof says about the structure of $\mathbb{R}^n$

Open sets in $\mathbb{R}^n$ (after compactification to S^n and product with $\mathbb{CP}^2$) carry:

1. **Controlled Poincaré–Lefschetz duality** at shrinking mesh, so algebraic Poincaré pairs exist with propagation $\rightarrow 0$.

2. **A degree-one invariant pinch** after Haar averaging of a small open tail, so the scalar L -class is the manifold germ of $S^n \times \mathbb{CP}^2$, not a multiple of p .
3. **No p -divisible integer signature tail** for that germ after transferring across a free C_p -space T built from local sheets.

In slogan form: $\mathbb{R}^n$ is **signature-tail rigid** under continuous p -adic orbit quotients. The rigidity is measured in $L_4(A_G(T))$, not on the possibly wild orbit space.

Q6 — Is the contradiction “orbit space factoring + wonky local sphere maps”?

Partly yes, but the wonky maps are controlled chain maps, not sphere maps of unbounded degree.

- Yes: the degree-one pinch f is H -invariant, so it factors through the orbit space of the tail. Finite quotients G_i act on chain models up to homotopy (graph type).
- The “wonky” part is compression to an exterior band (Lemma 10.1) and a duality-compatible homotopy idempotent E whose Karoubi corner forgets to m_f .
- The **basic contradiction** is arithmetic on integer signatures:

$$\sigma_{\text{tail}}(\alpha) = [(1, 1, 1, \dots)] \quad \text{and} \quad \sigma_{\text{tail}}(\alpha) \in p\mathbb{Z}^{\mathbb{N}}/\mathbb{Z}^{(\mathbb{N})},$$

hence $1 \in p\mathbb{Z}$.

Orbit-space factorization is necessary but not sufficient. The paradox is that factorization plus faithfulness (unbounded p -power finite quotients) plus Poincaré corners would force an integral class of signature 1 to be eventually p -divisible.

Section map

Sections	Role
1	Statement; two-part outline; conventions (rational chains, integral L -groups)
2	Asymptotic categories $A_G(X)$; support Karoubi filtrations
3	Equivariant signatures; Lemma 3.1 (cyclotomic); tail extraction σ_{tail}
4	Divisibility: local triviality of τ , nilpotence of ν , Theorem 4.3
5	Signs; compatible idempotent splitting; scalar normalization
6	Localization boundaries; union of Poincaré pairs
7	Acyclic carriers; dual-block Poincaré–Lefschetz at mesh η
8–9	Geometric data: chart, Haar pinch, detector T , chain models on $S^n \times \mathbb{CP}^2$

Sections	Role
10–11	Compression, graph corner, forgetting to m_f
12	Iterated boundary evaluation: $\Delta_1 m_f = \varepsilon[\mathbb{CP}^2]$
13	Realization of signature 1; proofs of 1.1 and 1.2

Audit ledger (Targets A–C)

Targeted audit ledger (Sherlock targets A–C)

Scope: line-by-line on load-bearing lemmas only (plan: targeted audit).

Grades: **PASS** = argument appears complete and correctly cited; **WEAK** = plausible but citation gap, omitted routine detail, hypothesis that needs expert/Lean check, or certificate not independently rerun; **FAIL** = identified break.

Overall targeted verdict: No FAIL found on the PDF’s load-bearing lemmas. Several **WEAK** items remain — especially **Target C** (Lean certificate vs published theorem), which this audit did **not** compile. The proof does **not** “prove too much” on known sponge models (see 03_sponge_stress_test.md).

Target A — Signature-tail lattice / integer signatures independent of p -power

The arithmetic engine is coarser than OpenAI’s $L_d^{-1}\mathbb{Z}$ lattice: ordinary signatures of rational symmetric forms are **integers**, and the tail map lands in $\mathbb{Z}^{\mathbb{N}}/\mathbb{Z}^{(\mathbb{N})}$. The “bounded denominator” claim is just integrality of σ , plus that two lifts of an $L_4(V_G)$ class agree eventually.

Lemma 3.1 — Cyclic arithmetic

PDF: §3, p. 3

Lean: AlgebraicReduction.lean, EquivariantSignature.lean

Grade: **PASS**

- If $G = C_{p^a} = \langle g \rangle$ and $\text{Sign}_G(V, b)(g) = 0$, then $p \mid \sigma(V, b)$.
- Write $\text{Sign}_G = \sum m_j \chi_j$; $F(t) = \sum m_j t^j$ is divisible in $\mathbb{Z}[t]$ by Φ_{p^a} , so $F(1) = \sigma$ is divisible by $\Phi_{p^a}(1) = p$.
- Standard cyclotomic fact; Lean proves the polynomial form (prime_dvd_eval_one_of_primitive_root) and the character form. No residual risk here.

Proposition 3.2 — Tail extraction

PDF: §3, pp. 3–4

Grade: **WEAK** (depends on negative K of $\mathbb{Q}[G]$)

- Semisimple rings $\mathbb{Q}[G_i]$ have vanishing negative K -groups [Wei13, III.4]; Bass’s polynomial definition commutes with the finite-support subcategory S , so $K_{-k}(S) = 0$ for $k \geq 1$.
- Rothenberg comparison $\Rightarrow L_j(S) = \bigoplus_i L_j^p(\mathbb{Q}[G_i])$. Karoubi filtration by finite index sets lifts every $L_4(V_G)$ class to $L_4(P)$; two lifts differ by $L_4(S)$, so coordinate classes agree eventually.
- **Why WEAK:** this is exactly the NegKHighAll / decoration-map bijectivity package that occupies most of HSFormal/LTheory/ (()104 modules). The PDF cites Weibel as a black box for $K_{-k}(\mathbb{Q}[G]) = 0$. If that comparison fails for the *asymptotic sequence category* (not just a single group ring), the tail map is not well-defined as a homomorphism to the product-modulo-sum. Lean claims to prove the iterated bounded-category form; this audit did not check those modules line-by-line.

Theorem 4.3 — Integer tail divisibility

PDF: §4, p. 4

Grade: PASS (with WEAK dependence on Lemma 4.2 and Prop. 3.2)

- Extract forms (V_i, b_i) . Nilpotence $\Rightarrow (p - \tau_i)^s[V_i, b_i] = 0$ in $L_4^p(\mathbb{Q}[G_i])$ for large i .
- Character of τ_i at a generator g_i vanishes, so $p^s \text{Sign}_{G_i}(V_i, b_i)(g_i) = 0$, hence that character is zero (complex numbers), and Lemma 3.1 gives $p \mid \sigma$.
- The arithmetic implication is clean (`character_vanishes_of_prime_power_annihilation` in Lean). Load-bearing gaps sit in the geometric nilpotence and in tail extraction, not here.

Proposition 13.1 / Prop. 12.4 — Realization of signature 1

PDF: §§12–13, pp. 15–16

Grade: WEAK

- After n localization boundaries, one component of $T \times S^0$ retains exactly one copy of $\mathbb{CP}^2$, sign $\varepsilon \in \{1, -1\}$ independent of i . Middle form $\langle 1 \rangle$; half-unit quadratic lift has signature 1.
- Identity-character trace identifies ordinary signatures of equivariant forms with those of their scalar images, so $\sigma_{\text{tail}}(\alpha) = [(1, 1, \dots)]$.
- **Why WEAK:** (i) sign bookkeeping through n cuts, one ordered diagonal, and component projection is the same class of foot-gun as OpenAI’s cylinder signs; (ii) “point control + subdivision identifies the constant model” is sketched rather than expanded; (iii) the half-unit comparison $\varphi/2$ on rational Poincaré complexes is standard Ranicki but must survive every exterior quotient. No contradiction found on reading; this is the geometric–algebraic interface of realization.

Target B — Transfer / averaging multiplies by p without leaving the integral lattice

Here “integral lattice” means: $\tau \otimes (-)$ and $p \cdot \text{id}$ act on the same ultimate L -groups of $A_G(T)$, and Haar averaging produces a **degree-one** (hence signature-one after $\mathbb{CP}^2$) class rather than a p -multiple.

Lemma 4.1 — Local triviality of τ

PDF: §4, p. 4

Grade: PASS

- On a compact $U \subset T/C_p$ inside a covering trivialization, a sheet function $c : L \rightarrow C_p$ gives an orthogonal equivariant isometry $\tau \otimes (-) \cong \text{id}^{\oplus p}$ of propagation zero.
- Compact positively separated sheets $\Rightarrow$ controlled morphisms eventually preserve them. Standard covering-space linear algebra. Appears complete.

Lemma 4.2 — Nilpotence $\nu^s = 0$

PDF: §4, p. 4

Grade: WEAK

- Finite compact cover of T/C_p of size s , independent of i . Induct via support Mayer–Vietoris from Lemma 2.2: $\nu = 0$ on $A \cap B \Rightarrow \partial(\nu x) = 0 \Rightarrow \nu x$ comes from $L_*(A_A) \oplus L_*(A_B)$; induction kills the first summand by ν^{s-1} , local triviality kills the second.
- **Why WEAK:** the argument assumes the support filtration of Lemma 2.2 induces a genuine localization / Mayer–Vietoris sequence in **ultimate** L -theory, with ν acting naturally on the sequence. Carlsson–Pedersen [CP95, Thm 4.2] is cited later for boundaries; here the MV sequence is called “natural” from Lemma 2.2 without repeating the $L\langle -\infty \rangle$ hypotheses (functoriality, additivity, Karoubi localization with connecting maps — listed as conventions on p. 2). Specialists should check that ν commutes with ∂ strictly enough for the nilpotence induction, including at $s = 1$.

Lemma 2.2 — Support calculus

PDF: §2, pp. 2–3

Grade: **PASS** (with residual on “equality means eventual equality”)

- Duality-invariant Karoubi filtration; $A_A \cap A_B = A_Y$; quotient equivalences; $A_G(S) \simeq A_S$; fully faithful when $d(S, Z) > 0$.
- Compactness of X is used to send labels approaching both A and B into Y . Relabeling by nearest-point projection has propagation $\rightarrow 0$.
- Residual: asymptotic categories identify morphisms only eventually. The filtration splits must be **uniform in the sequence index** for later nilpotence. The writeup addresses this via null sequences r_i and filtered orthogonal splits; looks carefully written.

Lemma 8.1 + Haar pinch (8.1)–(8.6)

PDF: §8, pp. 9–10

Grade: **PASS**

- Free point: nonzero closed subgroups of $\mathbb{Z}_p$ are the tails H_j ; if all stabilizers were nonzero, discrete $\mathbb{Z}$ would have finite orbits, hence finite image by [Rob75], contradicting effectivity.
- Haar average F_0 is continuous, invariant, $|F_0(x) - x| \leq 2D$; compactified pinch f has degree one by straight-line homotopy to the identity pinch s_ρ .
- Same geometric idea as OpenAI’s **lem:averaged-test**, done in coordinates rather than as a map $S^d \rightarrow Y \rightarrow S^d$. Finite $\mathbb{Z}/p$ cannot produce a sequence K_i with displacements $\rightarrow 0$ while remaining a proper open subgroup of a finite group. **Sanity:** see sponge doc.

Graph idempotent / Lemma 5.1–5.2 / Prop. 10.2

PDF: §§5, 10–11

Grade: **WEAK**

- Homotopy action of G_i on (C, φ) produces $E = q_i^{-1} \sum_g A_g \otimes R_{g^{-1}}$ with $E^2 \simeq E$ and $E\Phi \simeq \Phi E^*$ in the **exterior quotient**.
- Balmer–Schlichting [BS01] splits E in $K^b(\text{Kar } Q)$; normalized corner $\varphi_D = q^{-1} r \Phi r^*$. Scalar normalization (Lemma 5.2) says that after forgetting, the corner is isometric to (C, φ) .
- Compression (Prop. 10.2) kills defects through an exterior band around radius t ; detector cancellation $\chi(sg^{-1})\chi(g) = \chi(s)$ is the reason free sheets work.
- **Why WEAK:** (i) all homotopies, including products of $\sim q_i^2$ terms, are claimed to be uniformly controlled independent of rank — the paper emphasizes “uniform control rather than

a bound on the number of summands,” which is the right slogan but easy to under-prove; (ii) splitting “once in the sequence category” must produce asymptotically small representatives; (iii) signs in (5.1) and the double dual identification are spelled out, but the graph-type calculus of §9–10 is the longest formal stretch. Lean’s `CornerClass.lean` / `GraphCorner` is the matching formal target.

Target C — Lean certificate vs published theorem

Grade: WEAK (certificate not rerun; statement alignment looks good)

Claim (upstream README)	What this audit checked
<code>Challenge.lean</code> states classical HSC, proof sorry, Mathlib only	PASS. Boundaryless topological n -manifold; locally compact second-countable Hausdorff group; joint continuity; faithful action; conclusion C^∞ Lie in the given topology.
<code>Solution.lean = HSFormal.hilbertSmith n M G</code> <code>HSFormal.hilbertSmith</code> proved in <code>HilbertSmithNegK.lean</code> from <code>NegKHighAll</code>	PASS as text: same type signature, one-line call. Text PASS, compile unchecked. <code>negKHighAll := negKHighAll_of theoremB Reg.regKarStatement</code> , then <code>hilbertSmith_all.1</code> . Earlier comments in <code>HilbertSmithModel.lean</code> still describe <code>NegKHighAll</code> as “being formalized on the v4.35 port”; the terminal file <code>HilbertSmithNegK.lean</code> claims that induction is now a theorem.
Axioms used: <code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>	UNVERIFIED. This environment did not run <code>lake env lean / #print axioms</code> . A garbled third-party paste of the README listed <code>HilbertSmithNegK.lean</code> and even <code>HSproof.pdf</code> as axioms; the vendored README does not . Treat the three-axiom claim as a hypothesis until printed.
238 modules, import closure of <code>HilbertSmithNegK.lean</code>	PASS as a file count (<code>find HSFormal -name '*.lean' \ wc -l = 238</code>). Full tree not vendored.
Comparator / SafeVerify “solution is okay”	UNVERIFIED. Scripts live upstream under <code>verify/</code> ; not rerun (Linux Landlock, ()15 GB).

Claim (upstream README)	What this audit checked
PDF $\leftrightarrow$ Lean statement	Mostly PASS. Lean Challenge is boundaryless; paper Theorem 1.1 allows boundary and proves it by interior restriction. Lean also packages <code>HilbertSmithWithBoundary</code> and <code>PadicExclusionWithBoundary</code> in <code>Statement.lean</code>. Paper uses rational chain coefficients throughout; Lean builds a concrete <code>LowerLTheory</code> model. Residual: whether every PDF lemma is in the import closure, vs a weaker model with leftover <code>LowerLTheory</code> hypotheses. The terminal theorem <code>hilbertSmith</code> has no remaining hypothesis in the source text.

PDF vs Lean packaging. The PDF is 17 pages of controlled L -theory. The Lean tree is an engineering expansion: Newman, Gleason–Yamabe, Tao’s Lemma 7, a colimit model of lower L -theory, and a high negative- K induction. That is expected. The risk is not “too many files”; it is that a **sorry** or extra axiom hides in `LTheory/Model/NegK/High/` after the README’s axiom list was written.

Suggested axiom-print command (not run here):

```
lake env lean --stdin <<< 'import HSFormal.HilbertSmithNegK
#print axioms HSFormal.hilbertSmith'
```

Cross-cutting residual risks (not graded FAIL)

1. **Ultimate decoration** $L^{(-\infty)}$ is used everywhere; projective-to-ultimate maps and idempotent-completion invariance are listed as required properties rather than proved in the PDF.
2. **Signs** in Poincaré–Lefschetz dual blocks (Lemma 7.4) and iterated boundaries (Prop. 12.4).
3. **Homology-manifold edge** is outside the theorem’s hypotheses; see sponge doc.
4. **No independent #print axioms** in this session.

Suggested expert / Lean priorities

Priority	Object	Why
1	<code>#print axioms HSFormal.hilbertSmith</code> on a clean <code>lake build</code>	Distill C
2	Lemma 4.2 nilpotence vs [CP95] MV in ultimate L	Distill B

Priority	Object	Why
3	Prop. 3.2 tail extraction / NegKHighAll	Distill A/C
4	Prop. 10.2–11.6 graph corner + detector cancellation	Realization
5	Prop. 12.4 iterated boundary $\rightarrow \varepsilon[\mathbb{CP}^2]$	Signs / evaluation

Sanity check (finite $\mathbb{Z}/p$)

See `03_sponge_stress_test.md`. Failure is at the **unbounded tail** K_i with displacements $\rightarrow 0$: a finite group has no such sequence. Degree-one Haar averaging over a proper open subgroup also fails. Arithmetic of Theorem 4.3 is not reached. **Sanity PASS.**

Sponge stress test (Q5)

Sponge / non-manifold stress test (Dabrowski)

Goal: Can the argument be run on spaces where faithful continuous $\mathbb{Z}_p$ -actions **exist**, thereby “proving too much”? If every lemma’s hypotheses hold on such a space, that is a candidate gap. If a manifold-specific hypothesis fails early, the proof is correctly scoped.

Framing: A Menger sponge where Hilbert–Smith is false does **not** refute a theorem about manifolds. It is a probe for overclaiming.

Sanity: finite $\mathbb{Z}/p$ on $\mathbb{R}^2$

Step	Status
Chart / free point	OK for a free rotation (or fails faithfulness differently)
Unbounded tail $K_i = p^{\ell_i} \mathbb{Z}_p$ with displacements $\rightarrow 0$	FAILS (group is finite)
Haar degree-one pinch through a proper open subgroup	FAILS
L_4 tail divisibility / $\mathbb{CP}^2$ realization	Not reached

Why the tail fails. Realization needs a sequence of kernels $K_i \subset H' \subset H \cong \mathbb{Z}_p$ with $\ell_i \rightarrow \infty$, so that $G_i = H/K_i$ has unbounded p -power order while displacements on a chart ball tend to zero. A finite cyclic group has no such nested open subgroups. Separately, any invariant map through a free $\mathbb{Z}/p$ -orbit space has degree divisible by p (transfer), so there is no degree-one averaged pinch.

Conclusion: The proof does not incorrectly rule out finite cyclic actions. Break is at §8 geometric data, before Theorem 4.3. **Sanity PASS.**

Break-point matrix

For each model: first lemma/hypothesis that fails when one tries to copy the proof.

– means the step still makes sense; the chain stops at the first **BREAK**.

Model	Faithful $\mathbb{Z}_p$?	First BREAK	Later notes
Topological n -manifold (theorem setting)	Conjecturally no; paper claims no	(<i>none — target of proof</i>)	Full chain
Finite $\mathbb{Z}/p$ on $\mathbb{R}^2$	N/A (finite)	Unbounded tail K_i ; no deg-1 Haar pinch	Transfer degree

Model	Faithful $\mathbb{Z}_p$?	First BREAK	Later notes
$\mathbb{Z}_p \times \mathbb{R}^n$ (left trans- lation on $\mathbb{Z}_p$)	Yes	Not a manifold / no Euclidean chart (Lemma 8.1 chart)	Locally $\mathbb{Z}_p \times \mathbb{R}^n \not\approx \mathbb{R}^{n+k}$
p -adic solenoid Σ_p	Yes (standard free action; orbit S^1)	Not a manifold ; no chart ball in $\mathbb{R}^n$; no $S^n \times \mathbb{CP}^2$ from a coordinate compactification	Averaging to S^1 exists; Poincaré–Lefschetz dual blocks of Lemma 7.4 need a combinatorial manifold
Menger com- pactum μ^n (Dran- ishnikov 1989)	Yes — free action of any zero-dimensional compact group	Not a manifold ; Newman fails; no chart; Lemma 7.4 triangulation fails	Orbit dim can equal n ; irrelevant because chart already fails
Menger sponge / curve μ^1	Yes	Same as μ^n	Classic Cantor-like 1-dimensional universal curve
Raymond– Williams com- pactum	Yes; $\dim X = n$, $\dim X/\mathbb{Z}_p = n + 2$	Not a manifold ; chart / PL dual cells fail	Historically tied to Yang dim-raising; this paper does not use that obstruction
ENR homology n - manifold (Bryant– Ferry– Mio– Weinberger; Bing dogbone $\times \mathbb{R}$)	Unknown whether faithful $\mathbb{Z}_p$ can exist	OPEN EDGE	Local duality may partially survive; paper’s Lemmas 7.3–7.5 and 12.1–12.2 use combinatorial triangulations of manifolds. Homology manifolds need not be triangulable in this sense

Detailed sponge notes

1. Dranishnikov Menger actions (including the sponge)

Fact: Every Menger manifold (in particular μ^n) admits a free continuous action of $\mathbb{Z}_p$ (Dranishnikov, *Math. USSR Izv.* 32 (1989)).

Why the Dabrowski argument stops:

1. μ^n is not locally Euclidean $\Rightarrow$ Lemma 8.1's chart (image containing $B(0, 4)$) is undefined.
2. Haar coordinate average F_0 is an average of points in $\mathbb{R}^n$, not an abstract barycenter on a compactum.
3. $W = S^n \times \mathbb{CP}^2$ is built from the **compactified coordinate sphere**. There is no such sphere for a Menger chart.
4. Lemma 7.4 needs a finite combinatorial triangulation of a compact oriented manifold with dual blocks. Menger compacta are not manifolds.
5. Newman's theorem (used in Corollary 1.2, not in Theorem 1.1's realization) is a manifold theorem.

The divisibility half (Theorem 4.3) is stated for an abstract compact free C_p -space T . That half **could** be copied on a Menger free action, producing p -divisible tails in $L_4(A_G(T))$. The contradiction never fires because realization of signature 1 never starts. **“Proves too much?”** No.

2. Solenoid

The inverse-limit circle $\Sigma_p = \varprojlim (S^1 \xrightarrow{\times p} S^1)$ carries a free $\mathbb{Z}_p$ -action with orbit space S^1 .

- Orbit space is a manifold; the **total space** is not.
- One can average maps to circles, but the paper's degree-one pinch is a compactified Euclidean chart map $s_\rho \circ F_0$.
- Break: chart + $S^n \times \mathbb{CP}^2$ + dual-block Poincaré–Lefschetz.

3. Raymond–Williams

These examples show dim-raising for $\mathbb{Z}_p$ actions on **compacta**. The Dabrowski proof **does not use** Yang's cohomological-dimension relation. On RW spaces the Euclidean PL pipeline never starts. They neither confirm nor refute the L -theory steps; they only show that wild orbit spaces exist — which the paper already allows, since it never bounds $\dim Y$.

4. Homology manifolds (edge)

Topological manifolds $\Rightarrow$ ENR homology manifolds, but not conversely in high dimensions. Local Čech duality and orientation sheaves can exist without Euclidean charts.

Paper ingredient	Homology manifold?
Newman	Partial analogues exist; not automatic
Open $O \subset \mathbb{R}^n$ containing $B(0, 4)$	Fails without resolution/chart
Haar coordinate average	Needs vector space structure
Combinatorial dual blocks (Lemma 7.4)	Fails without a triangulation of a manifold
Abstract $L_4(A_G(T))$ divisibility	Could make sense for a free C_p -space T
$\mathbb{CP}^2$ evaluation	Needs the product Poincaré manifold $S^n \times \mathbb{CP}^2$

If a free $\mathbb{Z}_p$ action on an ENR homology manifold were found, one would check whether some substitute for Lemmas 7–12 still produces a signature-one tail. That is a **stress region**, not a present counterexample to the written theorem.

Shared Euclidean ingredients (why sponges fail the same way as OpenAI)

Both proofs need, in some packaging:

1. Newman rigidity (corollary / reduction).
2. Local Euclidean charts.
3. An integral signature (Witt sheaf vs L_4 tail / $\mathbb{CP}^2$).
4. Arbitrarily small open subgroups of Homeo (Haar averaging / unbounded finite quotients).

Sponges break (1)–(2) immediately, so they do not test (3)–(4). That is the correct scope for a manifold theorem. See `05_vs_openai.md`.

Public responses (Q3)

Public responses to the Dabrowski Hilbert–Smith preprint (Q3)

Search date: 2026-10-08

Paper: A. Dabrowski’s AI agents, *Integral tail signatures and the Hilbert–Smith conjecture*, 2026-10-08, in adbrw/HS_proof.

Executive finding

As of the morning of 2026-10-08 there is **no published technical rebuttal or confirmation** of this proof’s correctness. The repository is hours old (`created_at` 2026-10-07T08:08:15Z; paper commit 2026-10-08 09:06:46 UTC). Coverage, if any, is announcement-level. This Sherlock pass is therefore an independent first read, not a digest of consensus.

Lean status claimed by the author: a Lean 4 / Mathlib formalization of Corollary 1.2 (HSFormal/, 238 modules), with `Challenge.lean` / `Solution.lean` for comparator and SafeVerify. Claimed axioms: `propext`, `Classical.choice`, `Quot.sound`. **This audit did not rerun the checkers.**

Sources checked (2026-10-08)

Venue	Finding
GitHub adbrw/HS_proof	Public repo; 0 stars at vendor time; README states Thm 1.1 / Cor 1.2 and the Lean claims
MathOverflow	No hit on this preprint as of search (OpenAI Hilbert–Smith thread 515824 was closed as work-checking; unrelated to Dabrowski)
arXiv	No matching submission found under this title at search time; paper lives on GitHub as <code>HSproof.pdf</code>
Wikipedia / Tao 2011 blog / classical MO 32656	Background on the conjecture only; no 2026 Dabrowski discussion
Third-party HTML paste of the README	A garbled copy listed <code>HilbertSmithNegK.lean</code> and <code>HSproof.pdf</code> among “axioms.” The vendored README does not. Treat that paste as unreliable; settle the axiom list by <code>#print axioms</code>

What the upstream README claims (not independently verified)

- Lean v4.35.0-rc3; mathlib 1a547d8...; TauCeti c7af81f.
- `lake env lean axiom print` of `HSFormal.hilbertSmith`.
- Comparator last line: “Your solution is okay!”; SafeVerify: “SafeVerify check passed.”
- Resource needs: Linux ≥ 5.19 with Landlock, ()15 GB disk, 16 GB RAM.

Disclaimer in the PDF (p. 16)

The paper states that various LLMs (Anthropic, OpenAI, open-source) were used under human guidance at every stage; “the core idea and Lean4 formalization were obtained by Claude Opus 5.5 agents”; budget “below US\$1,000.” That is process context, not a correctness argument.

What to watch next

1. An independent `#print axioms HSFormal.hilbertSmith` from a clean lake build.
2. Expert notes from controlled topology / L-theory (Ranicki, Carlsson–Pedersen users) on §§2–6 and §§10–12.
3. Comparison with the OpenAI Witt-sheaf argument (same obstruction class; different packaging) — `05_vs_openai.md`.
4. Any MathOverflow / arXiv appearance under a non-announcement title.

Implication for this Sherlock session

Public silence on *technical* content means grades in `02_audit_ledger.md` are provisional expert-prep. Target C is **WEAK** primarily because the Lean certificate was not rerun here, not because a false axiom was found in the PDF.

Versus OpenAI

Dabrowski vs OpenAI: same obstruction class, different packaging

Both preprints claim the Hilbert–Smith conjecture in every finite dimension by excluding faithful continuous $\mathbb{Z}_p$ -actions on topological manifolds. Neither uses Yang’s orbit-dimension raising, and neither is Pardon’s 3-manifold mapping-class argument. They sit in the same **family**: integrality versus p -divisibility of a signature / Witt invariant manufactured from a Euclidean chart and a small open subgroup of Homeo.

OpenAI audit: docs/Hilbert_Smith_Audit.pdf (this repo, 2026-09-23 preprint).

Dabrowski audit: this directory.

One-line mechanisms

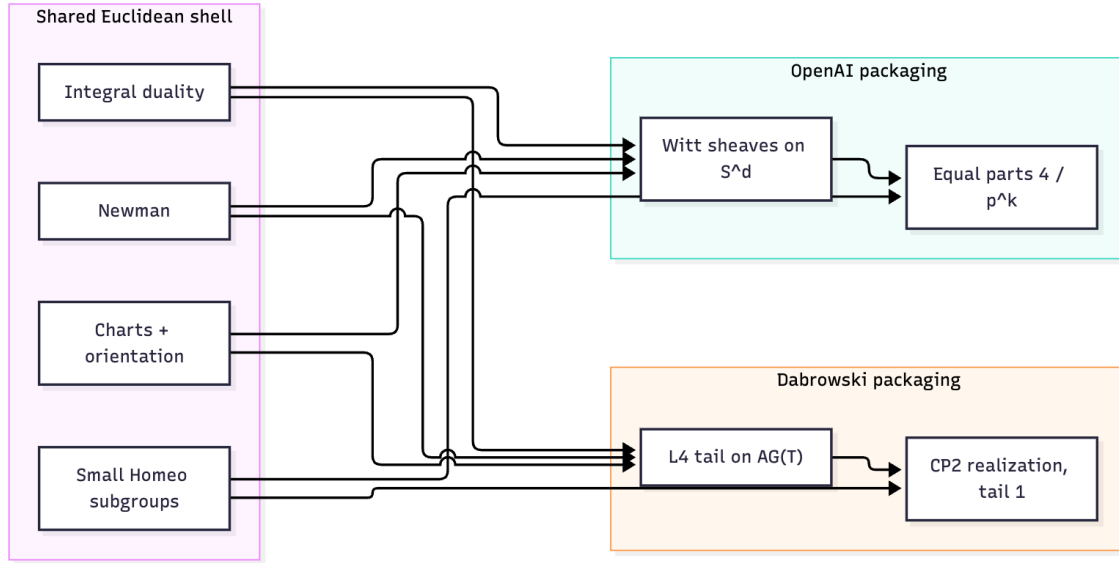
OpenAI (2026-09-23)	Dabrowski (2026-10-08)
Invariant sheaf Witt group $E_d(S^d)$ of continuous polyhedral forms	Ultimate quadratic L_4 of an asymptotic category $A_G(T)$
Integral fixed denominator lattice $L_d^{-1}\mathbb{Z}\bar{u}_d$, $L_d = 2^{e_d}$	Ordinary integer signatures; tail in $\mathbb{Z}^{\mathbb{N}}/\mathbb{Z}^{(\mathbb{N})}$
p -input p^k equal character pieces summing to $4\bar{u}_d$	Permutation form $\tau = \mathbb{Q}[G_i/P_i]$ locally p copies of id; $\nu^s = 0$
Contradiction $L_d^{-1}\mathbb{Z}$ for $p^k > 4L_d$	$1 \in p\mathbb{Z}$
Test Odd-dimensional sphere S^d after space stabilizing $O \subset \mathbb{R}^n$	$S^n \times \mathbb{CP}^2$; iterated boundaries recover $\mathbb{CP}^2$
Average Haar-average a degree-one pinch through $Y = W^+/H$	Haar-average coordinates in a chart; pinch f on the compactified S^n
Formal notation found in the OpenAI dump	Claimed Lean 4 certificate (HSFormal/, 238 modules) — not rerun here

Shared Euclidean ingredients

These four properties are what sponges lack and what both proofs spend their geometric budget on:

1. **Newman rigidity.** Small compact Lie actions that barely move a chart ball are trivial. Used by OpenAI in chart reduction; by Dabrowski in Corollary 1.2 (after p -adic exclusion) via [Pa19].
2. **Local Euclidean charts + orientation.** OpenAI: invariant $O \subset \mathbb{R}^n$, stabilize into S^d . Dabrowski: free point, chart containing $B(0, 4)$, compactify to S^n , product with $\mathbb{CP}^2$.
3. **Local duality / integral signature.** OpenAI: Verdier duality + Balmer triangular Witt + semialgebraic comparison. Dabrowski: controlled Poincaré–Lefschetz dual cells + Ranicki Poincaré pairs + Karoubi localization.
4. **Arbitrarily small open subgroups of Homeo.** Both need a tail of $\mathbb{Z}_p$ small enough that Haar averaging preserves degree one (OpenAI) or produces a signature-one $\mathbb{CP}^2$ germ (Dabrowski), and fine enough that finite quotients are unbounded p -powers.

On Menger / solenoid / Raymond–Williams spaces, (1)–(2) **fail first** for both arguments. That is why neither “proves too much.”



Where they diverge

Topic	OpenAI	Dabrowski
Dimension shift	Odd stabilize $d > n$ so $\pi_*(S^d)$ is Serre-rationally concentrated	Keep dimension n ; add $\mathbb{CP}^2$ to land in L_4 after n boundaries
Character theory	Finite partitions of p -power roots of unity on the punctured target	Equivariant signature characters of $\mathbb{Q}[G_i]$; cyclotomic Lemma 3.1
Lattice	Uniform 2-power denominator from $\mathcal{P} \rightarrow \mathcal{T}$ comparison	No denominator bookkeeping; integers from rational Witt / L_4
Orbit space	Explicit test $S^d \rightarrow Y \rightarrow S^d$	Invariant pinch on S^n ; orbit space of T is a covering, not the manifold quotient
Cited black boxes	Balmer–Walter 2002; Woolf; Hardt	Carlsson–Pedersen 1995; Ranicki 1980/89/92/99; Balmer–Schlichting 2001; Weibel negative K
Weak spots (this repo’s grades)	Dense doubling; BW localization; character coproducts; equal-parts isometry	Tail extraction / NegK; MV nilpotence; graph corner; $\mathbb{CP}^2$ evaluation; Lean axiom print
Boundary	Statement is without boundary	Paper allows boundary; Lean has both forms

What a common research question looks like

If both arguments are pointing at the same rigidity of Euclidean space, the mathematical question is **not** “which PDF is correct?” but:

Which of (1)–(4) is the essential obstruction to continuous $\mathbb{Z}_p$ -actions, and which can fail independently?

That is the MathOverflow rewrite in `docs/POST_DRAFTS.md`: a question about Euclidean structure, with both preprints as motivation.

Audit status comparison (targeted, 2026-10-08)

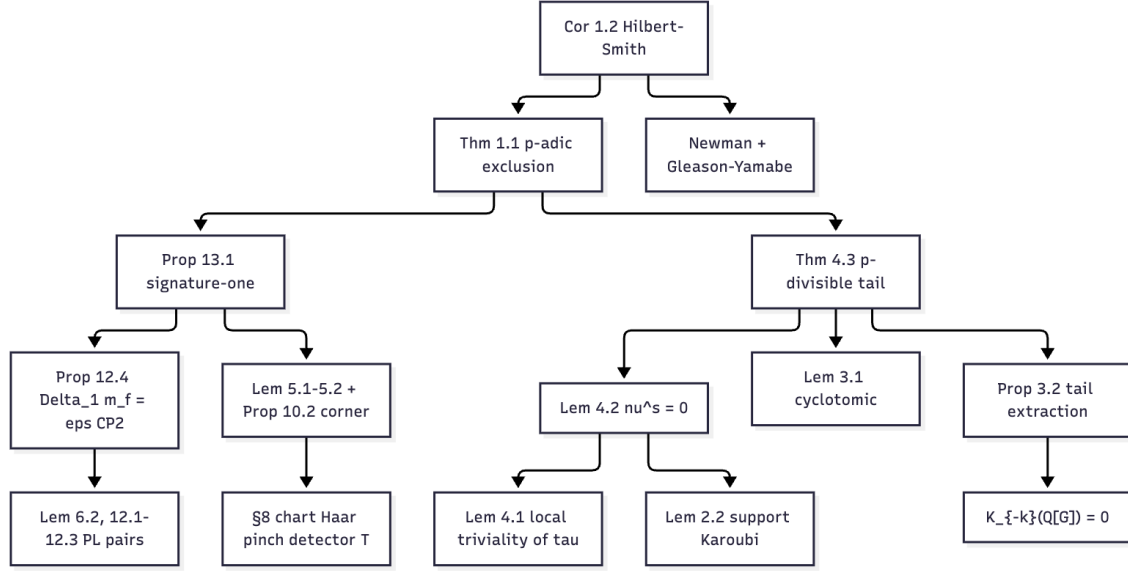
	OpenAI	Dabrowski
FAIL	0	0
Headline	4 lemmas	5 clusters (NegK/tail, MV nilpotence, graph corner, $\mathbb{CP}^2$ signs, Lean axioms)
WEAK		
Sponge	No	No
over-claim		
Finite	Breaks at averaged-test	Breaks at unbounded tail / deg-1 Haar
$\mathbb{Z}/p$		
sanity		
Lean	None	Claimed; not independently printed here

Lemma dependency ledger

Lemma dependency ledger (Dabrowski)

Extracted from the vendored PDF `HSprouf.pdf` (17 pp.) and sparse Lean headers. There is no public LaTeX tree; labels below are PDF lemma numbers.

Core dependency chain (contradiction)



Targeted audit nodes (by Sherlock target)

Target	PDF objects	Sparse Lean
A	Lem 3.1, Prop 3.2, Thm 4.3, Prop 12.4, Prop 13.1	<code>AlgebraicReduction.lean</code> , <code>EquivariantSignature.lean</code> , <code>HilbertSmithNegK.lean</code>
B	Lem 2.2, 4.1, 4.2; §8 Haar pinch; Lem 5.1–5.2; Prop 10.2	<code>ChartData.lean</code> , <code>CharacterDetector.lean</code> , <code>CornerClass.lean</code> , <code>ControlledDuality.lean</code>
C	Challenge/Solution vs Thm 1.1 / Cor 1.2	<code>Challenge.lean</code> , <code>Solution.lean</code> , <code>Statement.lean</code> , <code>HilbertSmith*.lean</code>
Sanity	Unbounded K_i ; deg-1 Haar	<code>ChartData.lean</code> comments on displacement $D < 1/64$

External citations used as black boxes

Citation	Role in proof
[CP95] Carlsson–Pedersen	Karoubi localization, connecting maps, pair description of boundaries
[Ran80I], [Ran80II] [Ran89], [Ran92]	Algebraic Poincaré pairs and unions Additive / lower L -theory; ultimate decoration; $L(Q) \cong L(\text{Kar } Q)$
[Ran99]	Controlled Poincaré–Lefschetz, dual cells
[BS01] Balmer–Schlichting	Idempotent splitting in the bounded derived category of Karoubi completion
[Wei13]	$K_{-k}(\mathbb{Q}[G]) = 0$ for finite G
[Pa19] Pardon survey	Standard reduction; Uniform Newman
[Gle52], [Yam53]	No small subgroups $\Rightarrow$ Lie
[Rob75]	Pointwise finite families $\Rightarrow$ finite image (free point)
[Hat02]	Simplicial approximation; degree classifies maps $S^n \rightarrow S^n$ for $n \geq 1$

Lean engineering (not in the 17-page PDF)

Module cluster	Role
Inputs/GY, Inputs/Lemma7, Newman/	Gleason–Yamabe, Tao 2011 Lemma 7, Newman prime order
LTheory/Model/ including NegK/High/	Concrete lower L -theory; NegKHighAll
Brouwer/	Five MIT-licensed files (upstream README)

Sherlock session log

Sherlock session — Dabrowski Hilbert–Smith preprint

Date: 2026-10-08

Workspace: /Users/ajjc/proj/Hilberts-5th-0AI

Protocol: Sherlock Mode (6 steps) + plan dabrowski_hs_sherlock

Step 1 — candidate failure sources (wide net)

1. Newman / chart reduction (same family as OpenAI `prop:chart-reduction`).
2. Integral tail construction: L_4 / Witt / Karoubi / Ranicki Poincaré pairs.
3. p -divisibility / transfer / averaging on finite covers of T .
4. Negative- K / `NegKHighAll` / decoration-map bijectivity.
5. Lean vs PDF gap: `sorry`, extra axioms, statement mismatch.
6. Homotopy / duality signs (cylinder / dual-block foot-guns).
7. Overclaim on sponges/solenoids if hypotheses are weaker than manifolds.

Step 2 — distill (most likely)

- **A.** Signature-tail lattice / integer signatures independent of p -power.
- **B.** Transfer/averaging multiplies by p without leaving the integral setting.
- **C.** Lean certificate actually proves the published theorem.

Step 3 — investigate

- Vendored `HSproof.pdf` (SHA-256 `e305fd47...`) and sparse Lean from https://github.com/adbrw/HS_proof at `abb0c44`.
- Line-by-line on load-bearing PDF lemmas (not all 238 Lean modules).
- Same sponge matrix as OpenAI; comparison table in `05_vs_openai.md`.

Step 4 — analysis

- **Mechanism:** integer signature tail: realization gives $\sigma_{\text{tail}} = 1$; divisibility gives eventually $p \mid \sigma$; hence $1 \in p\mathbb{Z}$. Same obstruction class as OpenAI, packaged as Ranicki $L_4 + \mathbb{CP}^2$ rather than Witt sheaves on S^d .
- **Audit (A–C):** 0 FAIL on PDF load-bearing lemmas. WEAK: Prop. 3.2 / NegK, Lemma 4.2 MV nilpotence, graph corner §§10–11, Prop. 12.4 signs, Lean axiom print.
- **Sanity:** finite $\mathbb{Z}/p$ breaks at unbounded tail / degree-one Haar.
- **Sponges:** Menger/solenoid/RW break at chart / PL dual cells; no overclaim. Homology manifolds OPEN EDGE.
- **Public:** no technical rebuttal as of 2026-10-08 morning. Garbled third-party README paste listed `HSproof.pdf` as an axiom; vendored README does not.

Step 5 — follow-ups

1. Independent `#print axioms HSFormal.hilbertSmith` after `lake build`.
2. Expert check of Lemma 4.2 against [CP95] ultimate L Mayer–Vietoris.
3. Expert check of Prop. 12.4 iterated boundary evaluation.
4. MO rewrite as Euclidean-structure question citing both proofs (not work-checking).

Step 6 — cleanup

/tmp/hs_dabrowski clone may be deleted after confirming `dabrowski/` is complete. No temporary diagnostic code was added to the proof. Full `HSFormal/` (238 modules) and `HilbertSmithLean4.zip` were **not** copied into the repo on purpose (sparse vendor).

Deliverables

- `dabrowski/PROVENANCE.md`, `dabrowski/paper/`, `dabrowski/lean/` (sparse)
- `dabrowski/docs/01-05`, lemma ledger, this log
- `dabrowski/docs/Dabrowski_HS_Audit.pdf`
- Root `README.md` two-proof index; `docs/POST_DRAFTS.md` MO rewrite

Appendix: contents

Contents

Proof overview (Q1, Q2, Q4, Q6)	2
Proof overview — Dabrowski Hilbert–Smith preprint (2026-10-08)	2
Statement	2
Mechanism in one paragraph	2
Q1 — Why no p-adic homeomorphisms near id?	4
Q2 — Geometric or algebraic?	4
Q4 — What the proof says about the structure of $\mathbb{R}^n$	4
Q6 — Is the contradiction “orbit space factoring + wonky local sphere maps”?	5
Section map	5
Audit ledger (Targets A–C)	7
Targeted audit ledger (Sherlock targets A–C)	7
Target A — Signature-tail lattice / integer signatures independent of p -power	7
Lemma 3.1 — Cyclic arithmetic	7
Proposition 3.2 — Tail extraction	7
Theorem 4.3 — Integer tail divisibility	8
Proposition 13.1 / Prop. 12.4 — Realization of signature 1	8
Target B — Transfer / averaging multiplies by p without leaving the integral lattice	8
Lemma 4.1 — Local triviality of τ	8
Lemma 4.2 — Nilpotence $\nu^s = 0$	8
Lemma 2.2 — Support calculus	9
Lemma 8.1 + Haar pinch (8.1)–(8.6)	9
Graph idempotent / Lemma 5.1–5.2 / Prop. 10.2	9
Target C — Lean certificate vs published theorem	10
Cross-cutting residual risks (not graded FAIL)	11
Suggested expert / Lean priorities	11
Sanity check (finite $\mathbb{Z}/p$)	12
Sponge stress test (Q5)	13
Sponge / non-manifold stress test (Dabrowski)	13
Sanity: finite $\mathbb{Z}/p$ on $\mathbb{R}^2$	13
Break-point matrix	13
Detailed sponge notes	14
1. Dranishnikov Menger actions (including the sponge)	14
2. Solenoid	15
3. Raymond–Williams	15
4. Homology manifolds (edge)	15
Shared Euclidean ingredients (why sponges fail the same way as OpenAI)	16
Public responses (Q3)	17
Public responses to the Dabrowski Hilbert–Smith preprint (Q3)	17
Executive finding	17
Sources checked (2026-10-08)	17
What the upstream README claims (not independently verified)	17

Disclaimer in the PDF (p. 16)	18
What to watch next	18
Implication for this Sherlock session	18
Versus OpenAI	19
Dabrowski vs OpenAI: same obstruction class, different packaging	19
One-line mechanisms	19
Shared Euclidean ingredients	19
Where they diverge	20
What a common research question looks like	21
Audit status comparison (targeted, 2026-10-08)	21
Lemma dependency ledger	22
Lemma dependency ledger (Dabrowski)	22
Core dependency chain (contradiction)	22
Targeted audit nodes (by Sherlock target)	22
External citations used as black boxes	22
Lean engineering (not in the 17-page PDF)	23
Sherlock session log	24
Sherlock session — Dabrowski Hilbert–Smith preprint	24
Step 1 — candidate failure sources (wide net)	24
Step 2 — distill (most likely)	24
Step 3 — investigate	24
Step 4 — analysis	24
Step 5 — follow-ups	24
Step 6 — cleanup	25
Deliverables	25
Directory README	28
Dabrowski Hilbert–Smith reverse verification	28
Repository README	29
Hilbert–Smith reverse verification	29
OpenAI (2026-09-23)	29
Dabrowski (2026-10-08)	30
MathOverflow / posts	30

Directory README

Dabrowski Hilbert–Smith reverse verification

Sherlock-mode audit of A. Dabrowski’s 2026-10-08 preprint *Integral tail signatures and the Hilbert–Smith conjecture*, with a sparse vendor of the claimed Lean 4 certificate.

Author: Cursor(Opus 5.5/Grok), driven by: Alan Coppola

Upstream: https://github.com/adbrw/HS_proof

Vendored at: `abb0c44d58d711235f20b666ef422a4ea87b9add`

PDF SHA-256: `e305fd474909694af1e032ab863d26411f5e9fd9c38e7001f5fac14fe19e5e75`

Main deliverable: `docs/Dabrowski_HS_Audit.pdf`

Rebuild (needs `pandoc + tectonic`): `python3 scripts/build_dabrowski_audit_pdf.py`

This directory sits **alongside** the OpenAI audit (`paper/`, `docs/`). Comparison: `docs/05_vs_openai.md`.

Repository README

Hilbert–Smith reverse verification

Sherlock-mode audits of two 2026 preprints claiming the Hilbert–Smith conjecture in every finite dimension. Both arguments sit in the same family — **integrality versus p -divisibility of a signature / Witt invariant** after a Euclidean chart and Haar averaging — not Yang dim-raising and not Pardon’s 3-manifold mapping-class theorem.

Author: Cursor(Opus 5.5/Grok), driven by: Alan Coppola

Proof	Date	Packaging	Audit PDF
OpenAI	2026-09-23	Witt/signature lattice on S^d ; equal parts $4/p^k$	<code>docs/Hilbert_Smith_Audit.pdf</code>
Dabrowski	2026-10-08	Integral tail in $L_4(A_G(T))$; $\mathbb{CP}^2$ realization vs $p\mathbb{Z}$	<code>dabrowski/docs/Dabrowski_HS_Audit.pdf</code>

Comparison: `dabrowski/docs/05_vs_openai.md`.

OpenAI (2026-09-23)

Upstream: <https://github.com/openai/math> (vendored at `adc7f1241b42e322a6451854ab7e4b4c146bf78a`)

Quick answers

Q	Answer
1. Near-id exclusion	Small open $\mathbb{Z}_p$ subgroups + Haar averaging give a degree-1 test through the orbit space; finite groups cannot.
2. Geometric vs algebraic	Mostly Witt/signature lattice arithmetic; thin geometric shell (Newman, chart, Serre).
3. Public responses	Process news only (2026-10-07); no technical rebuttal; no Lean for this proof.
4. Structure of $\mathbb{R}^n$	Signature-rigid under continuous $\mathbb{Z}_p$ orbit factorization after odd stabilization into S^d .
5. Menger sponges	Theorem false for sponges; proof breaks at chart/Newman — correctly scoped, does not overclaim.
6. Orbit factoring?	Necessary but not the whole story; contradiction is equal parts vs fixed denominator lattice.

Targeted audit: no FAIL; several WEAK (dense doubling, Balmer–Walter bridge, character coproducts, equal-parts isometry).

- `paper/` — vendored LaTeX + PDF + provenance
- `docs/01_proof_overview.md` — Q1, Q2, Q4, Q6
- `docs/02_audit_ledger.md` — lemma grades
- `docs/03_sponge_stress_test.md` — Q5 + sanity

- docs/04_public_responses.md — Q3
- docs/lemma_dependency_ledger.md
- Rebuild: python3 scripts/build_audit_pdf.py

Interactive canvas (open beside chat):
 hilbert-smith-audit.canvas.tsx

Dabrowski (2026-10-08)

Upstream: https://github.com/adbrw/HS_proof (vendored at abb0c44d58d711235f20b666ef422a4ea87b9add)

Quick answers

Q	Answer
1. Near-id exclusion	Small open tails of $\mathbb{Z}_p$ Haar-average a degree-1 pinch in a chart; unbounded finite quotients feed an L_4 tail of signature 1, which divisibility forces into $p\mathbb{Z}$.
2. Geometric vs algebraic	Mostly ultimate quadratic L -theory (Karoubi / Ranicki); thin Euclidean shell (chart, PL dual cells, $\mathbb{CP}^2$).
3. Public responses	None technical as of 2026-10-08 morning; claimed Lean 4 certificate not rerun here.
4. Structure of $\mathbb{R}^n$	Signature-tail rigid: a Euclidean germ realizes $\mathbb{CP}^2$ with $\sigma = 1$, which cannot be eventually p -divisible.
5. Menger sponges	Breaks at chart / combinatorial dual cells — correctly scoped.
6. Orbit factoring?	Necessary but not sufficient; contradiction is $\sigma_{\text{tail}} = 1$ vs eventual p -divisibility.

Targeted audit: no FAIL on load-bearing PDF lemmas; WEAK on NegK/tail extraction, Mayer-Vietoris nilpotence, the graph corner, $\mathbb{CP}^2$ evaluation, and Lean `#print axioms`.

- dabrowski/PROVENANCE.md
- dabrowski/paper/ — vendored HSproof.pdf + Challenge/Solution + lake files
- dabrowski/lean/ — sparse load-bearing modules (not the full 238)
- dabrowski/docs/01_proof_overview.md
- dabrowski/docs/02_audit_ledger.md
- dabrowski/docs/03_sponge_stress_test.md
- dabrowski/docs/04_public_responses.md
- dabrowski/docs/05_vs_openai.md
- Rebuild: python3 scripts/build_dabrowski_audit_pdf.py

MathOverflow / posts

Paste-ready Euclidean-structure rewrite (not work-checking): docs/POST_DRAFTS.md.